

$$\overset{1}{\varphi_n(\psi_n)}$$

$$\overbrace{= \gamma}$$

(*) key step $= \sum_{n \in \mathbb{Z}} \int_{\mathbb{R}} f\left(\begin{bmatrix} 2^n & b \\ 0 & 1 \end{bmatrix}\right) \frac{1}{2^n} \underbrace{\left| \omega_{\begin{bmatrix} 2^n & b \\ 0 & 1 \end{bmatrix}}(2^n \gamma) \right|}_{= \omega_2(1) = 1} db$

$$= \sum_{n \in \mathbb{Z}} \frac{1}{2^n} \int_{\mathbb{R}} f\left(\begin{bmatrix} 2^n & b \\ 0 & 1 \end{bmatrix}\right) db$$

Notice that

$$\text{Ad} \begin{bmatrix} 2^n & b \\ 0 & 1 \end{bmatrix} \gamma = \begin{bmatrix} 2^n & b \\ 0 & 1 \end{bmatrix} \begin{bmatrix} 0 & 1 \\ 0 & 0 \end{bmatrix} \begin{bmatrix} 2^{-n} & -2^{-n}b \\ 0 & 1 \end{bmatrix}$$

$$= \begin{bmatrix} 0 & 2^n \\ 0 & 0 \end{bmatrix} \begin{bmatrix} 2^{-n} & -2^{-n}b \\ 0 & 1 \end{bmatrix}$$

$$= \begin{bmatrix} 0 & 2^n \\ 0 & 0 \end{bmatrix} = 2^n \gamma$$

So $\det \text{Ad} \begin{bmatrix} 2^n & b \\ 0 & 1 \end{bmatrix} = 2^n$. This is not unimodular.

Representation Theory:

Let G be a matrix Lie group. V , a $\mathbb{C}$ -normed vector space.

A representation of G on V is a homomorphism

$$\rho: G \rightarrow GL(V) = \left\{ T: V \rightarrow V \mid \begin{array}{l} T \text{ is linear, cts} \\ \text{and invertible} \end{array} \right\}$$

Fact: T^{-1} is cts.

$$[\rho(g)^{-1} = \rho(g^{-1})]$$

$$\|Tx\| \leq \|T\| \|x\|$$

Recall:

$$T: V \rightarrow V, T \in GL(V)$$

(i) T is cts

$$(ii) \|T\| =$$

$$\sup \{ \|Tv\| : v \in V, \|v\| = 1 \}$$

$$< \infty$$

Such that for any v in V , $\rho \mapsto \rho(g)v : G \rightarrow V$ is cts.

$$[G \times V \rightarrow V : (g, v) \mapsto \rho(g)v \text{ is cts}]$$

(i) If V is finite-dim'l, then all norms on V are equivalent.

If $\beta = (v_1, \dots, v_n)$ is a basis for V , then for $\rho \in G$,

$$[Py]_\beta = [P_{ij}(y)] \quad , \quad \text{each } \underbrace{P_{ij}: G \rightarrow \mathbb{C}}_{\text{"matrix coeff fn."}}$$

Then P is cts $\Leftrightarrow$ all $P_{ij}: G \rightarrow \mathbb{C}$ are cts.

[usual notation of cty]

(ii) Consider $C_c(G)$, ($\mathbb{C}$ -valued) is G -normed space, with

$$\|f\|_\infty = \max_{j \in G} |f(j)| \quad . \quad \text{Define}$$

$$\lambda: G \rightarrow GL(C_c(G)) \quad \text{by} \quad \lambda(y)f = f \circ L_{y^{-1}}$$

$$(L_{y^{-1}}(h) = y^{-1}h) \quad . \quad \text{Notice} \quad \text{if } j, h \in G,$$

$$\begin{aligned} \lambda(yh)f &= f \circ L_{(yh)^{-1}} = f \circ L_{h^{-1}} \circ L_{y^{-1}} = \lambda(y)(f \circ L_{h^{-1}}) \\ &= \lambda(y)\lambda(h)f \end{aligned}$$

$$\text{Also, } \|\lambda(y)f\|_\infty = \|f\|_\infty, \text{ i.e., } \|\lambda(y)\| = 1.$$

Lemma: let $f \in C_c(G)$, $\varepsilon > 0$. Then there is a nbhd V of 1 in G so $\|f \circ L_g - f\|_\infty < \varepsilon$.

Conclay: $\lambda: G \rightarrow GL(C_c(G))$ is cts.

Pf: if $\lim_{n \rightarrow \infty} g_n = g$ in G , then for $f \in C_c(G)$,

$$\|\lambda(g_n)f - \lambda(g)f\|_\infty = \|\lambda g^{-1}(\lambda g_n f - \lambda g f)\|_\infty$$

$$= \|\lambda g^{-1}g_n f - f\|_\infty$$

$$= \|f \circ L_{g_n^{-1}g} - f\|_\infty \xrightarrow{n \rightarrow \infty} 0$$

As $g_n^{-1}g$ eventually enters each V from the lemma.

□

Wed, Mar. 5th

Pf (lemma): Fix an open nbhd V_0 of 1 in G

So $\bar{V}_0$ is compact.

if necessary, replace V_0 by $V_0 \cap V_0^{-1}$, so $V_0^{-1} = V_0$

($V_0^{-1} = \iota(V_0)$, $\iota(g) = g^{-1}$ is a Homeo)

Then

$V_0 \text{supp}(f) = \bigcup_{h \in \text{supp}(f)} R_h(V_0)$ is open ($R_h(g) = gh, \text{Homeo}$)

and

$(g, h) \mapsto gh : G \times G \rightarrow G$, allows $K = \bar{V}_0 \text{supp}(f)$ to be
seen to be compact.

Hence we have compact K with $\text{supp}(f) \subseteq K^\circ$

For each h in K , let W_h be an open nbhd of h

$$\text{s.t. } h' \in W_h \Rightarrow |f(h') - f(h)| < \frac{\varepsilon}{3}$$

city of multi'n on G provides for each h in K

open nbhds V_h of I , U_h of h , so $V_h U_h \subseteq W_h$

Compactness gives

$$K \subseteq \bigcup_{j=1}^n U_{h_j}$$

so $V = \bigcap_{j=1}^n V_{h_j} \cap V_0$ is open.

let $h \in K$, $j \in \{1, \dots, n\}$ be so at least one of the points

$$h, g_h \in \text{supp}(f) \quad [\text{o.w. notice that } f \circ \gamma(h) = 0 = f(h)].$$

Then there is h_j so $h \in U_{h_j} \subseteq W_{h_j}$ so $g_{h_j}, g_h \in W_{h_j}$

$$\begin{aligned}
 S_0 \quad |f(h) - f(yh)| &\leq |f(h) - f(hj)| + |f(hj) - f(jh)| + |f(jh) - f(yh)| \\
 &< \varepsilon/3 + \varepsilon/3 + \varepsilon/3 = \varepsilon.
 \end{aligned}$$

Thus. $\|f \circ g - f\|_\infty < \varepsilon$.

□.

(iii) if $\mathbb{C}$ -v.s. V admits an inner product, we call

$V = (V, (\cdot, \cdot))$ a Euclidean space.

let $\|v\| = (v, v)^{\frac{1}{2}}$. If $\rho: G \rightarrow GL(V)$ is a rep'n.

we call it a unitary rep'n if $(\rho(g)v, \rho(g)w) = (v, w)$, $g \in G$, $v, w \in V$.
[unitary isomery].

Unitary operators
We let $U(V) = \left\{ u \in GL(V) : (uv, uw) = (v, w) \text{ for } v, w \in V. \right\}$
 $u^{-1} \in GL(V)$

Euclidean structure on $C_c(G)$.

If $f, f' \in C_c(G)$, let

$$(f, f') = \int_G f \overline{f'} |w| \quad (\text{L.I. integral})$$

$$\|f\|_2 = (f, f)^{\frac{1}{2}} = \left(\int_G |f|^2 |w| \right)^{\frac{1}{2}}$$

[Ex: if $f \neq 0$ in $C_c(G)$, then $(f, f) = \int_G |f|^2 |w| > 0$]

Note: $g \in G$, $f, f' \in C_c(G)$

$$(\lambda_g f, \lambda_g f') = \int_G (f \overline{f'}) \circ \lambda_g |w|$$

$$= \int_G f \overline{f'} |w| = (f, f')$$

So each $\lambda_g \in U(C_c(G))$.

Lemma: if $f \in C_c(G)$, $g \mapsto \lambda_g f$ is etc in $\|\cdot\|_2$.

Pf: let $g_n \rightarrow g$ in G . Find h in $C_c(G)$ so

$h \geq 0$, $h|_K = 1$ where K is a compact set with

$g \operatorname{supp}(f) \in K^\circ$. Then

$$\|\lambda_{g_n} f - \lambda_g f\|_2^2 = \int_G |f \circ \zeta_{g_n^{-1}} - f \circ \zeta_{g^{-1}}|^2 |w|$$

$$\stackrel{\substack{n \\ \text{large} \\ \text{enough}}}{\Rightarrow} \int_G h |f \circ \zeta_{g_n^{-1}} - f \circ \zeta_{g^{-1}}|^2 |w|$$

$$\leq \|f \circ \zeta_{g_n^{-1}} - f \circ \zeta_{g^{-1}}\|_\infty^2 \int_G h |w|$$

$$\xrightarrow{n \rightarrow \infty} 0.$$

Def'n: A rep'n $\pi: G \rightarrow GL(V)$ $\left(\begin{array}{l} G\text{-matrix loc group} \\ GL(V) \text{ invertible } \mathbb{C}\text{-v.s.} \\ g \mapsto \pi(g)v \text{ l.s. } v \in V \end{array} \right)$

is called irreducible if a closed space $W \subseteq V$

being π -invariant implies that $W = \{0\}$ or V .

Schur's Lemma: let G be a (matrix) group,

$\pi: G \rightarrow GL(V)$, $\sigma: G \rightarrow GL(W)$ be any H.K. reps,

where V, W are irreducible finite dimensional.

(i) if $A: V \rightarrow W$ is linear and satisfies

$$A\pi(g) = \sigma(g)A \text{ for } g \in G \text{ (linearity)}$$

then either A is invertible or $A=0$.

(ii) If A intertwines π with itself, then $A = \lambda I$, $\lambda \in \mathbb{C}$.

Pf: (i) if $v \in \ker A$, then $A\pi(g)v = \sigma(g)Av = 0$, so $\ker A$ is π -inv.

if $w = Av \in \text{ran } A$, $\sigma(g)w = \sigma(g)Av = A\pi(g)v \in \text{ran } A$.

So $\text{ran } A$ is π -inv.

Then either $\ker A = \{0\}$, hence $\text{ran } A = W$,

or $\ker A = V$, hence $\text{ran } A = \{0\}$.

(ii). Let λ be any eigenvalue of A . Then $\lambda I - A$

intertwines π with itself. This is not invertible.

$$\Rightarrow \lambda I - A \Rightarrow$$

□.

Conclay: if G is abelian, any irreducible $\mathbb{C}$ -rep'n. is 1-dim'l.

Pf: Let $\sigma: G \rightarrow GL(W)$ be f.d. irred. Then for $g \in G$,

$\sigma(g)$ intertwines σ with itself, so $\sigma(g) = \lambda(g)I$, $\lambda(g) \in \mathbb{C}$.

Then $\lambda(yh)I = \sigma(yh) = \sigma(y)\sigma(h) = (\lambda(y)I)(\lambda(h)I) = \lambda(y)\lambda(h)I$

So $\lambda: G \rightarrow GL(\mathbb{C}) \cong \mathbb{C}^*$ is a HM.

If $\dim_{\mathbb{C}} V > 1$, we would find many σ -inv. subspaces. $\square$

Remark: Schur's lemma (i) holds for $\mathbb{R}$ -v.s. But (ii) may not.

Let $\pi: \mathbb{R} \rightarrow GL_2(\mathbb{R})$. $\pi(t) = \begin{bmatrix} \cos t & \sin t \\ -\sin t & \cos t \end{bmatrix}$ is real,

but not 1-dim'l.

Ex: (i) We can compute all f.d. irrep of $\mathbb{R} \cong \left\{ \begin{bmatrix} 1 & t \\ 0 & 1 \end{bmatrix} : t \in \mathbb{R} \right\}$

If $\chi: \mathbb{R} \rightarrow \mathbb{C}^* \subseteq GL_1(\mathbb{C})$ is a HM. One-parameter subgroup. Then

provides z in $\mathbb{C}$, so $\chi(t) = e^{zt}$. Notice that $z = \frac{d}{dt} \chi(t) |_{t=0}$

So if $z \neq z'$, $\chi_z(t) = e^{zt}$, $\chi_{z'}(t) = e^{z't}$ are distinct.

Hence f.d. irrep's are $\{\chi_z : z \in \mathbb{C}\} \cong \mathbb{C}$.

(ii). Let us consider $\mathbb{T} = U(1)$.

Notice that $t \mapsto e^{it} : \mathbb{R} \rightarrow \mathbb{T}$ is a covering map.

Kernel is $2\pi\mathbb{Z}$.

Now if $\chi : \mathbb{T} \rightarrow \mathbb{C}^\times$, let $\tilde{\chi}(t) = \chi(e^{it})$.

$$\Rightarrow \tilde{\chi}(t) = e^{zt}, \quad z = x + iy, \quad x, y \in \mathbb{R}.$$

But $\ker \tilde{\chi} \supseteq 2\pi\mathbb{Z}$. Hence for $n \in \mathbb{Z}$,

$$1 = e^{z \cdot 2\pi n} = e^{x \cdot 2\pi n} \cdot e^{i \cdot 2\pi n y} \Rightarrow x=0, y \in \mathbb{Z}.$$

Say $y = m$.

Conclusion, $\chi(e^{i2\pi t}) = e^{i2\pi m t}$, i.e., $\chi(w) = w^m$, $w \in \mathbb{T}$.

So finite dim'l irrep's of T are

$$\{ \chi_m : m \in \mathbb{Z} \}, \quad \chi_m(w) = w^m.$$

Monday, Mar. 17th.

Weyl's Thm: $\mathfrak{g}$ is a semisimple linear algebra.

$\pi: \mathfrak{g} \rightarrow \mathfrak{gl}(V)$ a f.d. rep'n. Then

if $W \subseteq V$ is π -invariant, then W admits a

π -invariant complement, W' , so $W \cap W' = \{0\}$.

$W + W' = V$, $\pi(\mathfrak{g})W' \subseteq W'$.

Towards concrete rep'n Thm.

G - (connected) compact matrix Lie group, $G \subseteq \mathrm{GL}_n(\mathbb{C})$.

Unitarization: We may assume $G \subseteq \mathrm{U}(n)$ (Otherwise, we replace

G by $SGS^{-1} \subseteq \mathrm{U}(n)$ for some $S \in \mathrm{GL}_n(\mathbb{C})$).

Hence $\mathfrak{g} = \text{Lie}(G) \subseteq \mathfrak{u}(n) = \{X \in \mathfrak{gl}_n(\mathbb{C}) : X^* = -X\}$
 (skew-adjoint / skew-Hermitian).

Remark: $\mathfrak{g} \cap i\mathfrak{g} = \{0\}$. Indeed, if $X \in \mathfrak{u}(n)$, with $iX \in \mathfrak{u}(n)$,

$$\text{then } X^* = -X \text{ while } (iX)^* = -iX^* \Rightarrow X = X^* = -iX$$

$$\Rightarrow X = 0.$$

$$\text{Hence } \mathfrak{g}_{\mathbb{C}} = \underbrace{\mathfrak{g} + i\mathfrak{g}}_{\cong \mathfrak{g} \otimes_{\mathbb{R}} \mathbb{C}} \text{ has } \dim_{\mathbb{C}} \mathfrak{g}_{\mathbb{C}} = \dim_{\mathbb{R}} \mathfrak{g}$$

Ex: (i) $\mathfrak{u}(n)_{\mathbb{C}} = \mathfrak{gl}_n(\mathbb{C})$, One inclusion is obvious.

if $X \in \mathfrak{gl}_n(\mathbb{C})$, then

$$X = \underbrace{\frac{1}{2}(X - X^*)}_{\in \mathfrak{u}(n)} - \frac{i}{2} \underbrace{i(X + X^*)}_{\in \mathfrak{u}(n)} \in \mathfrak{u}(n) + i\mathfrak{u}(n) = \mathfrak{u}(n)_{\mathbb{C}}.$$

(ii). $\mathfrak{su}(n)_{\mathbb{C}} = \mathfrak{sl}(n, \mathbb{C})$ [same proof as (i)].

(iii). $\mathfrak{so}(n)_{\mathbb{C}} = \mathfrak{so}(n, \mathbb{C}) := \{X \in \mathfrak{gl}(n, \mathbb{C}) : X^T = -X\}$ (skew-symmetric).

Recall: $\mathfrak{gl}(n, \mathbb{C}) = M_n(\mathbb{C})$, has inner product

$$(X, Y) = \text{Tr}(XY^*)$$

(i) Restricts to a $\mathbb{R}$ -inner product on $\mathfrak{gl}(n, \mathbb{R}) \times \mathfrak{gl}(n, \mathbb{R})$

$$(X, Y) = \text{Tr}(XY^T)$$

(ii). Now if $X, Y \in \mathfrak{u}(n)$ so $iX, iY \in \mathfrak{Her}_n(n)$. Also,

$$XY^* = (iX)(iY)^* = (iX)^*(iY) = X^*Y.$$

So

$$\begin{aligned}(X, Y) &= \text{Tr}(XY^*) = \text{Tr}(X^*Y) = \text{Tr}((XY^*)^*) = \overline{\text{Tr}(XY^*)} \\ &= \overline{(X, Y)}\end{aligned}$$

$\Rightarrow (X, Y) \in \mathbb{R}$. i.e., the usual inner product restricts to

$u(u) \times u(u)$ to be a $\mathbb{R}$ -inner product.

(iii). $g \in G \subseteq U(n)$, then $g^{-1} = g^*$. If $x, y \in \mathfrak{g}_{\mathbb{C}}$, then

$$(\text{Ad}(g)x, y) = \text{Tr}(g x g^* y^*) = \text{Tr}(x g^* y^* g)$$

$$= \text{Tr}(x (g^* y^* g)^*)$$

$$= (x, \text{Ad}(g^{-1})y)$$

$\Rightarrow \text{Ad}(g)^{-1} = \text{Ad}(g)^*$ on $\mathfrak{g}_{\mathbb{C}}$ (naturally, extend domain of $\text{Ad}(g)$).

However, we see on $\mathfrak{g}$ (natural domain of $\text{Ad}(g)$), that

$$\text{Ad}(g)^{-1} = \text{Ad}(g)^T \Rightarrow \text{Ad} : G \rightarrow O(\mathfrak{g}) = \{u \in L(\mathfrak{g}) :$$

$$(uX, uY) = (X, Y), \\ \forall X, Y \in \mathfrak{g}\}$$

Prop'n: $\mathfrak{g} (= \text{Lie}(G))$, G compact matrix group, is reductive.

Pf: Like in proof of Unitarization Thm (ii), we can show

$$\text{that } \mathfrak{g} = \bigoplus_{k=1}^m \mathcal{V}_k.$$

where $\mathcal{V}_k \perp \mathcal{V}_j$, $k \neq j$, and each $\mathcal{V}_k$ is $\text{Ad}(G)$ -invariant and

$\text{Ad}(\cdot)|_{\mathcal{V}_k}$ is irreducible. Hence, each $\mathcal{V}_k$ is $\text{ad}_\mathfrak{g}$ -invariant

($d(\text{Ad}) = \text{ad}$), so is an ideal in $\mathfrak{g}$, which is

either 1-dim'l, or itself a simple Lie algebra.

□

Let G be a connected compact matrix group, $\mathfrak{g} = \text{Lie}(G)$.

Def'n: A maximal torus is a maximal connected abelian subgp $T \subseteq G$

ie, if $T \subseteq T'$, where T' is connected & abelian, then

$$T = T'.$$

A Cartan subalgebra of $\mathfrak{g}$ is a maximal abelian lie subalgebra, $\mathfrak{t}$.

Prop'n: A max'l torus T is char'd, hence $T \cong \mathbb{T}^n$.

Also, $\mathfrak{t} = \text{Lie}(T)$, then $\mathfrak{t}$ is a Cartan subalgebra.

Finally, if $\mathfrak{t}$ is a Cartan subalgebra of $\mathfrak{g}$, +

$T = \exp(\mathfrak{t})$ is a maximal torus.

Pf: Since G is a topological group (Gts, multi'n, inverses), we see

that $\overline{T}$ must be an abelian gp, which is also connected.

Thus, $T = \overline{T}$.

Hence by A. 1. & 2, $T \cong \mathbb{T}^n$ for some n , and $\exp: \text{Lie}(T) \rightarrow T$

is a covering map.

if $\mathfrak{t} = \text{Lie}(T)$, if $\mathfrak{t} \subseteq \mathfrak{t}'$, $\mathfrak{t}'$ abelian, let

$T' = \overline{\exp(\mathfrak{t}')}$, so T' is abelian with $T \subseteq T'$. So

$T = T'$. Hence $\mathfrak{t} = \mathfrak{t}'$.

if $\mathfrak{t}$ is a Cartan subalgebra, $T = \overline{\exp(\mathfrak{t})}$. Then $\mathfrak{t} \subseteq \text{Lie}(T)$,

where the latter is abelian, so $\mathfrak{t} = \text{Lie}(T)$. As above.

$\exp: \mathfrak{t} \rightarrow T$ is a covering map.

□.

Examples: (i) $U(n)$. $T = \left\{ \begin{bmatrix} z_1 & & 0 \\ & \ddots & \\ 0 & & z_n \end{bmatrix} : z_i \in \mathbb{T} \right\} \cong \mathbb{T}^n$.

$\mathfrak{u}(n)$, $\mathfrak{t} = \text{span}_{\mathbb{R}} \{H_1, \dots, H_n\}$, each $H_k = i E_{kk}$.

obvious: $\exp(\mathfrak{t}) = T$, and $\text{Lie}(T) = \mathfrak{t}$.

Recall that $\mathfrak{su}(n)$ has basis

$$\{H_1, \dots, H_n, X_{ij} = \bar{E}_{ij} - \bar{E}_{ji}, Y_{ij} = i(\bar{E}_{ij} + \bar{E}_{ji}) : k=1, \dots, n, 1 \leq i < j \leq n\}$$

Then, for example,

$$[H_i, X_{ij}] = Y_{ij}, [H_j, Y_{ij}] = X_{ij}$$

So we see that $\mathfrak{t}$ is maximal abelian, by testing against linear combinations.

Notice by unitary diagonalization, if $g \in U(n)$, then for some $u \in U(n)$,

$$u g^* u \in \mathfrak{T}. \text{ Hence } \bigcup_{u \in U(n)} u \mathfrak{T} u^* = U(n).$$

$$(ii) \cdot SU(n), \mathfrak{T} = \left\{ \begin{bmatrix} z_1 & & 0 \\ & \ddots & \\ 0 & & z_n \end{bmatrix} : z_i \in \mathbb{T}, \prod z_i = 1 \right\} \cong \mathbb{T}^{n-1}$$

$$\mathfrak{su}(n), \mathfrak{t} = \mathfrak{span}_{\mathbb{R}} \{H_1 - H_2, \dots, H_{n-1} - H_n\}.$$

(iii) $SO(n)$, $n \geq 2$.

$$n=2m, \quad T = \left\{ \begin{bmatrix} u_{\theta_1} & & 0 \\ & \ddots & \\ 0 & & u_{\theta_m} \end{bmatrix} : u_{\theta} = \begin{bmatrix} \cos \theta & \sin \theta \\ -\sin \theta & \cos \theta \end{bmatrix} : \theta_i \in \mathbb{R} \right\}$$

$$\mathcal{L} = \left\{ \begin{bmatrix} \theta_1 H & & 0 \\ & \ddots & \\ 0 & & \theta_m H \end{bmatrix} : \theta_i \in \mathbb{R}, H = \begin{bmatrix} 0 & 1 \\ -1 & 0 \end{bmatrix} \right\}$$

$$n=2m+1, \quad T = \left\{ \begin{bmatrix} u_{\theta_1} & & 0 \\ & \ddots & \\ 0 & & u_{\theta_m} & \\ & & & 1 \end{bmatrix} : \theta_i \in \mathbb{R} \right\}.$$

$$\mathcal{L} = \left\{ \begin{bmatrix} \theta_1 H & & 0 \\ & \ddots & \\ 0 & & \theta_m H & \\ & & & 0 \end{bmatrix} : \theta_i \in \mathbb{R} \right\}.$$

Note: $m = \lfloor \frac{n}{2} \rfloor$, $\mathcal{L} = \text{Span}_{\mathbb{R}} \{ H_k : k=1, \dots, m \}$, $H_k = E_{2k+1, 2k} - E_{2k, 2k+1}$, $k=1, \dots, m$.

Let's check maximality of T . Consider $n=2m$. $h \in SO(2m) \setminus T$.

Write $h = [H_{ij}]$, $i, j = 1, \dots, m$; each $H_{ij} \in M_n(\mathbb{R})$.

$$u_h = [u_{\theta_i} H_{ij}] \quad , \quad h_u = [H_{ij} u_{\theta_j}]$$

Sp: $H_{ij} \neq 0$. If $[H_{ij}, H] = 0$, $i \neq j$, pick $u \in T$ so

$$u_{\theta_i} \neq u_{\theta_j} \quad , \quad \text{so } u_h \neq h_u.$$

If $[H_{ij}, H] \neq 0$, we can use $\theta_i = \theta_j = \frac{\bar{1}_i}{2}$ above, so $u_h \neq h_u$.

let us show that each elt of $SD(n)$ is conjugate to one in T .

if $u \in SD(n)$, we have that there is $v \in SD(n)$.

$$v u v^* = \begin{bmatrix} u_{\theta_1} & & 0 \\ & \ddots & \\ 0 & & u_{\theta_m} \end{bmatrix}. \quad (*)$$

To do this, first note that eigenvalues of u are $z_1, \bar{z}_1, \dots$
 $z_m, \bar{z}_m, (1)$.

If $z_k \in \mathbb{T} \setminus \mathbb{R}$, then there is $w \in \mathbb{C} \setminus \{0\}$ so

$$Uw = z_k w \Rightarrow U\bar{w} = \overline{Uw} = \overline{z_k w} = \bar{z}_k \bar{w}, \text{ so } \bar{z}_k \text{ is an eigenvalue of } \bar{w}.$$

Let $x = \operatorname{Re}(w)$, $y = \operatorname{Im}(w)$. Then

$$0 = (w, \bar{w}) = (x+iy, x-iy) = \|x\|^2 - \|y\|^2 + i(x, y)$$

$\uparrow$
 $\mathbb{C}^n$ -inner prod.
 eigenvalue of different
 eigenvalues

$$\Rightarrow (x, y) = 0.$$

Write $z_k = e^{i\theta_k}$, then

$$Ux = \operatorname{Re}(Uw) = \operatorname{Re}(e^{i\theta_k}(x+iy)) = (\cos\theta_k)x - (\sin\theta_k)y$$

$$Uy = \dots = (\sin\theta_k)x + (\cos\theta_k)y$$

$\rightsquigarrow$
 writ. basis
 (x, y) (normalized)

$$\begin{bmatrix} \cos(-\theta_k) & \sin(-\theta_k) \\ -\sin(-\theta_k) & \cos(\theta_k) \end{bmatrix}$$

if $n \geq m+1$, replace v by $-v$ ($v \in O(n) \setminus SO(n)$)

if $n \geq m$, replace v by

$$\begin{bmatrix} 1 & 0 & & \\ & 1 & & \\ & & \ddots & \\ 0 & & & 1 \end{bmatrix} v, \quad \begin{bmatrix} 0 & 1 \\ 1 & 0 \end{bmatrix} v_0 \begin{bmatrix} 0 & 1 \\ 1 & 0 \end{bmatrix} = 4\theta.$$

Black Box: If G is connected compact matrix grp.

with maximal torus T , any g in G admits h in G so

$$hgh^{-1} \in T.$$

Hence, all maximal tori are conjugate in G , $G = \bigcup_{T \in \mathcal{T}} T$
max'l tori.

Corollary: $\exp: \mathfrak{g} \rightarrow G$ is surjective.

Pf: If $\mathfrak{t} = \text{Lie}(\bar{T})$, $\exp(\mathfrak{t}) = T$. If $hgh^{-1} \in T$,

$$hgh^{-1} = \exp(\lambda), \lambda \in \mathfrak{t}.$$

$$\text{So } g = \exp(\underbrace{\text{Ad}(h^{-1})\lambda}_{\in \mathfrak{g}}).$$

□.

Corollary: $T = C_G(T) = \{g \in G : gh = hg \text{ for all } h \in T\}$

Pf: Let $g \in C_G(T)$. Consider

$$C_G(g) = \{h \in G : gh = hg\} \supseteq T.$$

So $T \subseteq C_G(g)_0$ (connected component).

By the last corollary, $g = \exp(X)$, $X \in \mathfrak{g}$, and for $t \in \mathbb{R}$ ($t \in \mathbb{C}$),

$$\exp(zX) \in C_G(\exp(X))_0 = C_G(Y)_0.$$

Applying Black Box to $C_G(Y)_0$, there is h in $C_G(Y)_0$, so

$$hgh^{-1} \in I, \text{ but } hg = gh.$$

□.

Prop'n: $\widehat{\mathbb{T}}^n = \{ \chi_v : v \in \mathbb{Z}^n \}, \chi_v(z) = z^v = z_1^{v_1} \dots z_n^{v_n}.$

Pf: (Induction): For $n=1$, we see this.

Sps this holds for $n-1$ ($n > 1$), let $\chi: \mathbb{T}^n \rightarrow \mathbb{T} = \mathbb{C}^\times$ be a lts character, $\zeta \in \mathbb{T}$, $z \in \mathbb{T}^{n-1}$. Then:

$$\begin{aligned} \underbrace{\chi(z, \zeta)}_{\in \mathbb{T}^n} &= \chi((z, 0)(1, \zeta)) = \chi(z, 0) \chi(1, \zeta) \\ &= z^v \zeta^{v_n}, \quad v \in \mathbb{Z}^{n-1}, v_n \in \mathbb{Z}. \end{aligned}$$

Evidently, each $\chi_v \in \widehat{T}_n$, and if $v \neq \mu$ in $\mathbb{Z}^n$, then

$$\chi_v \neq \chi_\mu$$

□.

Weights and Roots:

Let G be a compact, connected matrix group ($G \subseteq U(n)$).

T is maximal torus, $\mathfrak{g}, \mathfrak{t}$ associated Lie algebra.

Prop'n: If $\pi: G \rightarrow U(V)$ is a f.d. unitary rep'n. Then

$$[\pi|_T]_\beta = \begin{bmatrix} \chi_1 & & 0 \\ & \ddots & \\ 0 & & \chi_n \end{bmatrix}, \chi_1, \dots, \chi_n \in \widehat{T} \text{ (univ. basis } \beta)$$

Pf: This is simply complex reducibility for rep'n of a compact abelian gp. □

Consequence: let π be as above, $d\pi: \mathfrak{g} \rightarrow u(V) = \{T \in L(V): T^* = -T\}$

Then for $H \in \mathfrak{t}$,

$$d\pi(H) = \begin{bmatrix} \alpha_1(H) & & 0 \\ & \ddots & \\ 0 & & \alpha_n(H) \end{bmatrix}, \text{ each } \alpha_k = d\chi_k: \mathfrak{t} \rightarrow u(1) = i\mathbb{R}$$

($\mathbb{R}$ -linear)

Def'n: let $\Delta(\pi) := \{\alpha_1, \dots, \alpha_n\}$ (weights of π)

and we get a weight decomposition:

$$V = \bigoplus_{\alpha \in \pi(H)} V_\alpha, \text{ where each } V_\alpha = \{x \in V: d\pi(H)x = \alpha(H)x \text{ for } H \in \mathfrak{t}\}.$$

This is an orthogonal decomposition.

Note: Each $\alpha: \mathfrak{t} \rightarrow i\mathbb{R}$ is $\mathbb{R}$ -linear. Extend $\alpha: \mathfrak{t}_{\mathbb{C}} \rightarrow \mathbb{C}$ by

$$\alpha(H + ik) = \alpha(H) + i\alpha(k), \quad H, k \in \mathfrak{t}; \text{ making } \alpha \text{ } \mathbb{C}\text{-linear.}$$

Def'n: Recall that $\text{Ad}: G \rightarrow U(\mathfrak{g}_{\mathbb{C}})$ is unitary.

So we define the roots of G by

$$\Delta(\mathfrak{g}_{\mathbb{C}}) = \Delta(\text{Ad}) \setminus \{0\}.$$

We thus get the root decomposition

$$\mathfrak{g}_{\mathbb{C}} = \mathfrak{g}_0 \oplus \bigoplus_{\alpha \in \Delta(\mathfrak{g}_{\mathbb{C}})} \mathfrak{g}_{\alpha} = \mathfrak{t}_{\mathbb{C}} \oplus \bigoplus_{\alpha \in \Delta(\mathfrak{g}_{\mathbb{C}})} \mathfrak{g}_{\alpha}$$

$$\mathfrak{g}_{\alpha} = \{X \in \mathfrak{g}_{\mathbb{C}} : [H, X] = \text{ad}(H)X = \alpha(H)X, \text{ for } H \in \mathfrak{t}\}.$$

Note that $\mathfrak{g}_0 = \bigcap_{H \in \mathfrak{t}} \ker \text{ad}(H) = \mathfrak{t}_{\mathbb{C}}.$

Thm: (i) if $\pi: G \rightarrow U(V)$ is a f.d. rep'n, $\alpha \in \Delta(\mathfrak{g}_{\mathbb{C}})$,

$\beta \in \Delta(\pi)$. Then $d\pi(\mathfrak{g}_{\alpha})V_{\beta} \subseteq V_{\alpha+\beta}.$

(ii). if $\alpha, \beta \in \Delta(\mathfrak{g}_{\mathbb{C}}) \cup \{0\}$, then

$$[\mathfrak{g}_{\alpha}, \mathfrak{g}_{\beta}] \subseteq \mathfrak{g}_{\alpha+\beta}.$$

(iii) if $Z(\mathfrak{g}) = \{0\}$, then $\text{span}_{\mathbb{C}} \Delta(\mathfrak{g}_{\mathbb{C}}) = \mathfrak{t}_{\mathbb{C}}^*$ ($\mathbb{C}$ -linear functionals)

Pf: (i) $d\pi: \mathfrak{g} \rightarrow \mathfrak{u}(\mathcal{V})$ $\mathbb{R}$ -linear, extend to $\mathbb{C}$ -linear $d\pi: \mathfrak{g}_{\mathbb{C}} \rightarrow \mathfrak{gl}(\mathcal{V})$

let $H \in \mathfrak{t}_{\mathbb{C}}$, $X \in \mathfrak{g}_{\alpha}$, $v \in \mathcal{V}_{\beta}$. Then

$$\begin{aligned} d\pi(H) d\pi(X) v &= d\pi(X) d\pi(H) v + [d\pi(H), d\pi(X)] v \\ &= d\pi(X) \underbrace{\beta(H) v}_{v \in \mathcal{V}_{\beta}} + \underbrace{d\pi([H, X]) v}_{d\pi \text{ is derivation}} \end{aligned}$$

Note $[H, X] = \alpha(H) X$, as $X \in \mathfrak{g}_{\alpha}$

$$\text{So above} = (\alpha(H) + \beta(H)) d\pi(X) v$$

□.

(ii): Special case of (i).

(iii): let $H \in \mathfrak{t}_\mathbb{C}$. Then

$$H \in \bigcap_{\alpha \in \Delta(g_\mathbb{C})} \ker \alpha \Leftrightarrow \operatorname{ad}(H) = 0 \oplus \bigoplus_{\alpha \in \Delta(g_\mathbb{C})} \underbrace{\alpha(H)}_0 = 0$$

$$\Leftrightarrow H \in \mathfrak{z}(g_\mathbb{C})$$

But $\mathfrak{z}(g)_\mathbb{C} = \mathfrak{z}(g_\mathbb{C})$ (easy exercise), so $H=0$ in case above.

If $\mathfrak{z}(g) = \{0\}$, then $\bigcap_{\alpha \in \Delta(g_\mathbb{C})} \ker \alpha = \{0\} \Leftrightarrow \operatorname{span}_\mathbb{C} \Delta(g_\mathbb{C}) = \mathfrak{t}_\mathbb{C}^*$.

□.

ex: (i) $\mathfrak{u}(n)$.

$$\mathfrak{u}(n)_\mathbb{C} = \mathfrak{gl}_n(\mathbb{C}) = \bigoplus_{i,j=1}^n \mathbb{C} E_{ij}, \text{ where } \mathfrak{t}_\mathbb{C} = \bigoplus_{i=1}^n \mathbb{C} E_{ii}.$$

let $\varepsilon_i = (0, \dots, \underset{i\text{-th position}}{1}, \dots, 0) \in \mathbb{Z}^n$, and for $z = \begin{bmatrix} z_1 & & 0 \\ & \ddots & \\ 0 & & z_n \end{bmatrix} \in T$.

$$\text{Ad}(z) \bar{e}_{ij} = z_i \bar{z}_j \bar{e}_{ij} = z^{z_i - \bar{z}_j} \bar{e}_{ij}.$$

$$\text{let } e_i = dX_{\varepsilon_i}, \quad e_k \left(\begin{bmatrix} i\theta_1 & & \\ & \ddots & \\ 0 & & i\theta_n \end{bmatrix} \right) = i\theta_k.$$

So we end up seeing that roots are

$$\Delta(\mathfrak{u}(\mathfrak{u})_{\mathbb{C}}) = \{ \pm (e_i - e_j) : 1 \leq i < j \leq n \}$$

(ii) $SU(n)$, a similar computation yields

$$\Delta(\mathfrak{u}(\mathfrak{u})_{\mathbb{C}}) = \{ \pm (e_i - e_j) : 1 \leq i < j \leq n \}.$$

$\mathfrak{u}(\mathbb{C})$

Monday, Mar 24th, 2025.

let G be connected compact matrix gp, $\mathfrak{g} = \mathcal{L}_{\mathbb{C}}(G)$,
 $\in \mathcal{U}(m)$ $\in \mathcal{U}(m)$

$\mathcal{T}$, $\mathfrak{t}$ torus & Cartan.

Root Decomposition:

$$\mathfrak{g}_{\mathbb{C}} = \mathfrak{t}_{\mathbb{C}} \oplus \bigoplus_{\alpha \in \Delta(\mathfrak{g}_{\mathbb{C}})} \mathfrak{g}_{\alpha} - \mathfrak{g}_{\alpha} = \{X \in \mathfrak{g}_{\mathbb{C}} : [H, X] = \alpha(H)X, \forall H \in \mathfrak{t}_{\mathbb{C}}\}.$$

each $\alpha: \mathfrak{t} \rightarrow \mathbb{C} = i\mathbb{R}$ is $\mathbb{R}$ -linear.

extend to $\mathbb{C}$ -linear $\alpha: \mathfrak{t}_{\mathbb{C}} \rightarrow \mathbb{C}$.

$$\text{span}_{\mathbb{C}} \Delta(\mathfrak{g}_{\mathbb{C}}) = \underbrace{\overline{\mathcal{Z}(\mathfrak{g})}_{\mathbb{C}}}_{\text{linear functionals}}^{\perp} \in \mathfrak{t}_{\mathbb{C}}^*.$$

linear functionals while
vanish on $\mathcal{Z}(\mathfrak{g})_{\mathbb{C}} \subseteq \mathfrak{t}_{\mathbb{C}}$.

(ii). $SU(n)$, similar holds, $T = \left\{ \begin{bmatrix} z_1 & & 0 \\ & \ddots & \\ 0 & & z_n \end{bmatrix} : z_1, \dots, z_n \in \mathbb{T}, z_1 \cdots z_n = 1 \right\}$

$$\mathfrak{u}(n)_{\mathbb{C}} = \mathfrak{u}_n(\mathbb{C}) = \mathcal{L} \oplus \bigoplus_{1 \leq i < j \leq n} \mathbb{C} E_{ij} \oplus \mathbb{C} E_{ji}$$

$$\text{each } \mathbb{C} E_{ij} = \mathfrak{u}(n)_{\mathbb{C}} e_i - e_j, \quad i \neq j, \quad i, j \in \{1, \dots, n\}$$

$$\mathcal{L} = \text{span} \{H_1, \dots, H_{n-1}\}, \quad H_k = E_{kk} - E_{k+1, k+1}.$$

$\rightsquigarrow$ (formula for each e_j to be revisited).

Analysis on $SU(n)$ vs. $U(n)$:

$$\text{Note that } \mathfrak{u}(n) = \underbrace{i\mathbb{R} I}_{\mathfrak{z}(\mathfrak{u}(n))} \oplus \mathfrak{su}(n) \cong \underbrace{\text{Lie}(U(n) \times SU(n))}_{\cong \pi I}$$

let us consider $f: \mathbb{T} \times SU(n) \rightarrow U(n)$, $f(z, u) = zu$.

Surjectivity: if $u \in U(n)$, then $u = z \bar{z} u$, where $z^n = \det(u)$.

kernel: let $(z, u) \in \mathbb{T} \times SU(n)$.

$$zu = I \Leftrightarrow u = \bar{z} I \in \mathbb{T} \cap SU(n)$$

$$zI \in SU(n) \Leftrightarrow z^n = \det(zI) = 1 \Leftrightarrow z = \omega_n^k, k=0, \dots, n-1, \\ \omega_n = e^{\frac{2\pi i}{n}}$$

$$\text{Thus, } \ker f = \{(\omega_n^k, \omega_n^{-k} I) : k=0, \dots, n-1\} \cong \mathbb{Z}/n\mathbb{Z}.$$

(iii). $SO(n)$.

$$\text{Toy case: } n=2. \text{ let } C_2 = \frac{1}{\sqrt{2}} \begin{bmatrix} 1 & -i \\ i & 1 \end{bmatrix}, E_2 = \begin{bmatrix} 0 & 1 \\ 1 & 0 \end{bmatrix} \in U(2) \\ = E_2^*.$$

$$\text{with } C_2 C_2^T = E_2.$$

$$\text{let for } \theta \in \mathbb{R}. U_\theta = \begin{bmatrix} \cos \theta & \sin \theta \\ -\sin \theta & \cos \theta \end{bmatrix} \in SO(2).$$

$$C_2 U_\theta C_2^* = \frac{1}{2} \begin{bmatrix} 1 & -i \\ i & 1 \end{bmatrix} \begin{bmatrix} \cos \theta & \sin \theta \\ -\sin \theta & \cos \theta \end{bmatrix} \begin{bmatrix} 1 & 1 \\ i & -i \end{bmatrix}$$

$$= \begin{bmatrix} e^{i\theta} & 0 \\ 0 & e^{-i\theta} \end{bmatrix} := V_\theta.$$

$$\text{Also, } \bar{E}_2 V_\theta \bar{E}_2 = V_\theta^* = \overline{V_\theta}$$

$$n(\mathbb{C}) = \mathbb{R} \begin{bmatrix} 0 & 1 \\ -1 & 0 \end{bmatrix}, \quad C_2 \begin{bmatrix} 0 & 1 \\ -1 & 0 \end{bmatrix} C_2 = C_2 U_\theta C_2^* = \begin{bmatrix} i & 0 \\ 0 & -i \end{bmatrix} := \lambda,$$

$$\bar{\lambda} = \bar{E}_2 \lambda \bar{E}_2 = -\lambda^T.$$

Now sps $n > 2$.

$$n = 2m, \text{ let } C_{2m} = \frac{1}{\sqrt{2}} \begin{bmatrix} I_m & -iI_m \\ I_m & iI_m \end{bmatrix}.$$

$$\bar{E}_{2m} = \begin{bmatrix} 0 & I_m \\ I_m & 0 \end{bmatrix}, \quad F_{2m} = \begin{bmatrix} I_m & 0 \\ 0 & -I_m \end{bmatrix} \in U(2m)$$

involutions matrices.

$$n=2m+1, \text{ let } C_{2m+1} = \begin{bmatrix} C_{2m} & 0 \\ 0 & 1 \end{bmatrix}, \quad \bar{C}_{2m+1} = \begin{bmatrix} \bar{C}_{2m} & 0 \\ 0 & 1 \end{bmatrix}.$$

$$\bar{F}_{2m+1} = \begin{bmatrix} \bar{F}_{2m} & 0 \\ 0 & 1 \end{bmatrix} \in U(2m+1).$$

$$\text{Then } C_n C_n^T = \bar{E}_n, \quad C_n^T C_n = F_n.$$

$$\text{let } SO(\bar{E}_n) = \left\{ g \in SL_n(\mathbb{C}) : \underbrace{g^T \bar{E}_n g = \bar{E}_n, \bar{E}_n g \bar{E}_n = \bar{g}} \right\} \cong SU(n).$$

$$g^{-1} = \bar{E}_n g^T \bar{E}_n = (\bar{E}_n g \bar{E}_n)^T = \bar{g}^T = g^*$$

Then it is easy to see that

$$\text{Lie}(SO(\bar{E}_n)) := \mathfrak{so}(\bar{E}_n) = \left\{ X \in \mathfrak{gl}_n(\mathbb{C}) : \begin{aligned} &X^T \bar{E}_n + \bar{E}_n X = 0, \\ &\bar{E}_n X \bar{E}_n = -X \end{aligned} \right\}$$

$$= \left\{ X \in \mathfrak{gl}_n(\mathbb{C}) : \underbrace{-X^T = \bar{E}_n X \bar{E}_n = -X}_{\Rightarrow \text{Tr}(X) = 0} \right\}$$

let for $u \in \text{SD}(C_n)$, $\gamma(u) = C_n u C_n^*$. We shall see that

$$\gamma(\text{SD}(C_n)) = \text{SD}(\bar{C}_n).$$

let us see that $\gamma(\text{SD}(C_n)) \subseteq \text{SD}(\bar{C}_n)$:

$$\begin{aligned} \bar{C}_n \gamma(u) \bar{C}_n &= C_n C_n^T C_n u \underbrace{C_n^* C_n}_{I} C_n^T \\ &= \underbrace{C_n \bar{C}_n}_{\bar{C}_n} u C_n^T \\ &= \bar{C}_n u C_n^T \\ &= \overline{C_n u C_n^*} \quad , \quad u \text{ is real} \\ &= \overline{\gamma(u)} \end{aligned}$$

$$\begin{aligned} \gamma(u)^T \bar{C}_n \gamma(u) &= (C_n u C_n^*)^T C_n C_n^T C_n u C_n^* \\ &= \bar{C}_n u^{-1} \underbrace{C_n^T C_n C_n^T C_n}_{\bar{C}_n^2 = I} u C_n^* = \bar{C}_n C_n^* = \overline{C_n C_n^T} \\ &= \bar{\bar{C}_n} = \bar{C}_n \end{aligned}$$

Now let us see that $\gamma^{-1}(SO(\bar{E}_n)) \subseteq SO(U_n)$. Let $J \in SO(\bar{E}_n)$.

$$\gamma^{-1}(J)^T \gamma^{-1}(J) = (C_n^* J C_n)^T C_n^* J C_n$$

$$= \underbrace{C_n^T J^T \bar{C}_n C_n^* J C_n}_{= \bar{E}_n J^{-1} \bar{E}_n}$$

$$= C_n^T C_n C_n^T J^{-1} \underbrace{C_n C_n^T \bar{C}_n}_{= \overline{C_n^* C_n} = I} C_n^* J C_n$$

$$\underbrace{\quad}_{= I}$$

$$= I$$

$$= C_n^T C_n C_n^T C_n$$

$$= I_n^2 = I.$$

check that $\gamma^{-1}(SO(\bar{E}_n)) \subseteq GL_n(\mathbb{R})$.

Conclusion: $\gamma: SO(U_n) \rightarrow SO(\bar{E}_n)$ is an isomorphism. This

induces an isomorphism

$$d\gamma: \mathfrak{so}(n) \rightarrow \mathfrak{so}(\tilde{E}_n).$$

In $SO(n)$ ($n \geq m$), we have maximal torus

$$T = \left\{ h_z = \begin{bmatrix} z_1 & & & \\ & \ddots & & \\ & & z_m & \\ & & \bar{z}_1 & \\ & & & \ddots & \\ 0 & & & & \bar{z}_m \end{bmatrix} : z_1, \dots, z_m \in \mathbb{T} \right\}$$

(1) $\leftarrow$ if $n \geq m+1$

$$h_z = \gamma \left(\begin{bmatrix} u_0 & & \\ & \ddots & \\ 0 & & u_{0m} \end{bmatrix} \right), \quad z_1 = e^{i\theta_1}, \dots, z_m = e^{i\theta_m}.$$

[Note that a max'l torus in $SO(m)$ has dim m ,

and this is clearly a torus, thus max'l].

Similar holds for torus if $n \geq m+1$.

□.

Cartan: $\mathfrak{k} = \text{Lie}(\mathbb{T}) = \left\{ \begin{bmatrix} i\theta_1 & & & 0 \\ & \ddots & & \\ & & i\theta_m & \\ 0 & & & -i\theta_1 & \dots & -i\theta_m \end{bmatrix} : \theta_k \in \mathbb{R} \right\}$

$$= \text{Span}_{\mathbb{R}} \{ H_k = i(\bar{E}_{kk} - E_{k+m, k+m}) : k=1, \dots, m \}$$

We wish to compute $\Delta(\mathfrak{so}(n)_{\mathbb{C}})$; we will need to know

$\mathfrak{so}(\bar{E}_n)_{\mathbb{C}}$. Say $n=2m$:

if $X \in \mathfrak{so}(n)$, $X = \begin{bmatrix} A & B \\ C & D \end{bmatrix}$, $A, B, C, D \in M_m(\mathbb{C})$.

$$\bar{X} = \bar{E}_n X \bar{E}_n = -X^T$$

$$\begin{bmatrix} \bar{A} & \bar{B} \\ \bar{C} & \bar{D} \end{bmatrix} = \begin{bmatrix} D^T & C^T \\ B^T & A^T \end{bmatrix} = - \begin{bmatrix} A^T & C^T \\ B^T & D^T \end{bmatrix}$$

$$\Rightarrow \mathfrak{so}(\bar{E}_{2m}) = \left\{ \begin{bmatrix} A & B \\ \bar{B} & -A^T \end{bmatrix} : A \in M(m), B \in \mathfrak{so}(m, \mathbb{C}) \right\} \subseteq \mathfrak{u}(2m)$$

Exercise: $\dim_{\mathbb{R}} \mathfrak{so}(E_{2m}) = \dim_{\mathbb{R}} \mathfrak{so}(2m) = \frac{2m(2m-1)}{2}$

Then $\mathfrak{so}(E_{2m})_{\mathbb{C}} = \left\{ X \in \mathfrak{gl}_n(\mathbb{C}) : E_n X E_n = -X^T \right\}$

$$= \left\{ \begin{bmatrix} A & B \\ C & -A^T \end{bmatrix} : A \in \mathfrak{gl}_n(\mathbb{C}) = \mathfrak{u}(m)_{\mathbb{C}}, \right. \\ \left. B, C \in \mathfrak{so}(m, \mathbb{C}) \right\}$$

$$\dim_{\mathbb{C}} \mathfrak{so}(E_{2m}) = \frac{2m(2m-1)}{2}$$

Now we let h_2 be as above, $z \in \mathbb{T}^m$

$\mathbb{C}$ -Basis for $\mathfrak{so}(E_{2m})_{\mathbb{C}} = H_k = E_{kk} - E_{k+m, k+m}$, $k=1, \dots, m$
in $\mathfrak{L}$,

$i \neq j$ in $\{1, \dots, m\}$:

$$X_{ij} = E_{ij} - E_{j+m, i+m} \text{ in } \left\{ \begin{bmatrix} A & 0 \\ 0 & -A^T \end{bmatrix} : A \in \mathfrak{gl}_n(\mathbb{C}) \right\}$$

not in $\mathfrak{L}$.

$$Y_{ij} = E_{i,j+m} - E_{j,i+m} \in \left\{ \begin{bmatrix} 0 & B \\ C & 0 \end{bmatrix} : B, C \in \mathcal{O}(m, \mathbb{C}) \right\}.$$

Compute $\text{Ad}(h_z) X_{ij} = h_z X_{ij} h_z^{-1} = z_i \bar{z}_j X_{ij} \xrightarrow{\text{like in}} e_i - e_j$

$$\text{Ad}(h_z) Y_{ij} = h_z Y_{ij} h_z^{-1} = z_i \bar{z}_j Y_{ij} \xrightarrow{i \neq j \in [m]} e_i + e_j$$

$$\text{Hence } \Delta(\mathcal{O}(E_{2m})_{\mathbb{C}}) = \{ \pm(e_i - e_j), \pm(e_i + e_j) : 1 \leq i < j \leq m \}$$

where $e_i \in \mathfrak{t}_{\mathbb{C}}^*$, $e_i(H_j) = \delta_{ij}$, linear.

Now let $n = 2m+1$. As above,

$$\mathcal{O}(E_{n+1}) = \left\{ X = \begin{bmatrix} A & B & x \\ \bar{B} & -A^T & x^T \\ -x^T & 0 & 0 \end{bmatrix} : A \in \mathfrak{u}(m), B \in \mathcal{O}(m, \mathbb{C}), \right. \\ \left. x \in \mathbb{R}^{2m} \text{ (column)} \right\}$$

$\uparrow$
 $\text{Tr}(X) = 0$

$$\text{So } \mathcal{O}(E_{2m+1}) = \left\{ \begin{bmatrix} A & B & x^T \\ C & -A^T & x^T \\ -x & 0 & 0 \end{bmatrix} : A \in \mathfrak{gl}_m(\mathbb{C}), B, C \in \mathcal{O}(m, \mathbb{C}) \right\}$$

$x \in \mathbb{C}^{2m}$

Then to our previous basis, we add

$$Z_k = E_{k,2m+1} - E_{2m+1,k}, \quad k=1, \dots, 2m.$$

and

$$\text{Ad}(h_2) Z_k = \begin{cases} Z_k Z_k, & k \leq m \\ \bar{Z}_k Z_k, & k > m \end{cases} \rightsquigarrow \pm e_k, \quad k=1, \dots, m.$$

$$\text{Hence } \Delta(\mathfrak{so}(E_{m+1})_{\mathbb{C}}) = \left\{ \pm e_k, \pm(e_i - e_j), \pm(e_i + e_j) : \begin{array}{l} k=1, \dots, m \\ 1 \leq i < j \leq m \end{array} \right\}$$

Fact: All "simple" compact Lie group (really Lie algebra

with no non-trivial ideals):

$\mathfrak{su}(n)$, $\mathfrak{so}(2m)$, $\mathfrak{so}(2m+1)$, $\mathfrak{sp}(m)$, 3 classes of

"exceptional" Lie groups.

As before, G is a compact, connected matrix Lie group, $\mathfrak{g} = \text{Lie}(G)$

T max'l torus, $\mathfrak{t} = \text{Lie}(T)$.

Def'n: let $N_G(T) = \{g \in T : gTg^{-1} \subseteq T\}$ (= holds by looking at $\text{Ad}(G)|_{\mathfrak{t}}$).

The Weyl group is defined by

$$W = W(G) = N_G(T)/T.$$

If $w = gT$ in W , $H \in \mathfrak{t}$, then let $w \cdot H = \text{Ad}(g)(H)$
 $h \in T, w \cdot h = ghg^{-1}$ (well-defined)

$$(\text{span}_{\mathbb{C}} \mathfrak{s}(g_{\mathbb{C}})) = \mathfrak{s}(g_{\mathbb{C}})^{\perp} \subseteq \mathfrak{t}_{\mathbb{C}}^*$$

If $\lambda \in \mathfrak{t}^*$ ($\mathbb{R}$ -linear dual), let $w \cdot \lambda(H) = \lambda(w^{-1} \cdot H)$

and we extend to $\mathfrak{t}_{\mathbb{C}}^*$, hence restrict to

$$i\mathfrak{t}^* = \{\alpha: \mathfrak{t} \rightarrow i\mathbb{R}, \mathbb{R}\text{-linear}\}$$

Recall that $\Delta(\mathfrak{g}_{\mathbb{C}}) \subseteq i\mathfrak{t}^*$.

Thm: (i) The action of W on $\mathfrak{t}$ is faithful, hence

faithful on $\underbrace{\mathfrak{z}(\mathfrak{g}_{\mathbb{C}})^{\perp}}_{\text{annihilator of centre}} \subseteq \mathfrak{t}_{\mathbb{C}}^*$, hence on $\mathfrak{z}(\mathfrak{g}_{\mathbb{C}})^{\perp} \cap i\mathfrak{t}^*$.

(ii) if $\alpha \in \Delta(\mathfrak{g}_{\mathbb{C}}) \cup \{0\}$, $w = pT \in W$, then

$$\text{Ad}(g)g_{\alpha} \in g_{w \cdot \alpha}, \text{ so } w \cdot \alpha \in \Delta(\mathfrak{g}_{\mathbb{C}}) \cup \{0\}.$$

(iii). The action of W on $i\mathfrak{t}$ is faithful preserves

$\Delta(\mathfrak{g}_{\mathbb{C}})$, and is faithful on $\Delta(\mathfrak{g}_{\mathbb{C}})$

(iv). W is finite.

(v). For $h \in T$, $\{php^{-1} : p \in G\} \cap T = W \cdot T$

conjugate orbits.

Pf: (i) let $g \in N_G(T)$, then

$$\text{Ad } g|_{\mathfrak{t}} = I_{\mathfrak{t}} \Leftrightarrow g h g^{-1} = h, \text{ for } h \in T \quad (\text{ex } \alpha: \text{exp: } t \rightarrow \bar{t} \text{ covering map})$$

$$\Leftrightarrow g \in C_G(T)$$

$$\Leftrightarrow g \in T \quad (\text{ie, } C_G(T) = T).$$

Thus $w \cdot H = H$ for all $H \Leftrightarrow w = e \in T$ (identity coset)

$$\text{in } W = N_G(T)/T$$

Hence W acts faithfully, by adjoint action, on $\mathfrak{t}^*$, hence

on $\mathfrak{t}_{\mathbb{C}}^*$, and on $i\mathfrak{t}^*$.

We also note that as G acts trivially on $\mathcal{Z}(\mathfrak{g}_{\mathbb{C}})$,

by conjugation, the same must true of $N_G(T)$.

It follows W acts faithfully on $\mathfrak{z}(\mathfrak{g}_C)^\perp$, thus faithfully on $\mathfrak{z}(\mathfrak{g}_C)^\perp \cap i\mathfrak{t}^*$.

ii) let $H \in \mathfrak{t}$, $X \in \mathfrak{g}_\alpha$, $\alpha \in \Delta(\mathfrak{g}_C) \cup \{0\}$. Compute

$$\text{ad}(H) \text{Ad}(g) X = \text{Ad}(g) \underbrace{\text{Ad}(g^{-1}) \text{ad}(H) \text{Ad}(g)} X$$

$$= \text{Ad}(g) \underbrace{\text{ad}(\text{Ad}(g^{-1})H)}_{= w^{-1} \cdot H \in \mathfrak{t}} \underbrace{X}_{\in \mathfrak{g}_\alpha} \quad \text{ad} = d(\text{Ad})$$

$$= \text{Ad}(g) \alpha(w^{-1} \cdot H) X$$

$$= w \cdot \alpha(H) \text{Ad}(g) X$$

so $w \cdot \alpha \in \Delta(\mathfrak{g}_C) \cup \{0\}$, with $w \cdot \alpha \neq 0$ if $\alpha \neq 0$

(test with $w \cdot H$ in place of H with $\alpha(H) \neq 0$).

Moreover, $\text{Ad}(g)g_\alpha \in \mathfrak{g}_{W\alpha}$, and, likewise, $\text{Ad}(g^{-1})g_{W\alpha} \in \mathfrak{g}_\alpha$.

so $\text{Ad}(g)g_\alpha = g_{W\alpha}$ [Fact: g_α is always 1-dim for $\alpha \in \Delta(\mathfrak{g}_\mathbb{C})$]

iii). We saw that W preserves $\Delta(\mathfrak{g}_\mathbb{C})$ in (ii). Since

$\text{span}_\mathbb{C} \Delta(\mathfrak{g}_\mathbb{C}) = \mathfrak{z}(\mathfrak{g}_\mathbb{C})^\perp$, and thus as W acts on

$\mathfrak{z}(\mathfrak{g}_\mathbb{C})^\perp$, it must act faithfully on $\Delta(\mathfrak{g}_\mathbb{C})$.

iv). W acts faithfully on a finite set. ($W \hookrightarrow \text{Sym}(\Delta(\mathfrak{g}_\mathbb{C}))$)

v). Sp. $g \in G$ act satisfy that $ghg^{-1} \in T$.

Then $T \in C_G(ghg^{-1})$. If $h' \in T$, then

$$gh'g^{-1}ghg^{-1} = gh'hg^{-1} = gh'h'g^{-1} = ghg^{-1}gh'h'g^{-1}.$$

so we also have $gTg^{-1} \in C_G(ghg^{-1})$. Since T is a

max'l torus in G , is a maximal torus in

$C_G(yhy^{-1})_0$, so there is $z \in C_G(yhy^{-1})_0$ so

$$zy^{-1}z^{-1} = T, \text{ i.e., } zy \in N_{C_G(yhy^{-1})_0}(T).$$

If $w = zyT$, then $w \cdot h = zghy^{-1}z^{-1} = ghg^{-1}$, as

$$z \in C_G(yhy^{-1})_0.$$

□

Ex: (i) $U(n)$. $T = \{h_z = \begin{bmatrix} z_1 & & 0 \\ & \ddots & \\ 0 & & z_n \end{bmatrix} : z_1, \dots, z_n \in \mathbb{T}\} \cong \mathbb{T}^n$.

If $g \in N_{U(n)}(T)$, then $ghzg^{-1}$ has same eigenvalues

as h_z so $ghzg^{-1} = \begin{bmatrix} z_{\sigma(1)} & & 0 \\ & \ddots & \\ 0 & & z_{\sigma(n)} \end{bmatrix}$, $\sigma \in \text{Sym}(n)$.

$[h_z$ is regular : $z_1, \dots, z_n$ are distinct].

if $\sigma \in \text{Sym}(n)$. let $P_\sigma = \sum_{i=1}^n E_{\sigma(i), i}$; (permutation matrix),

and we have $ghg^{-1} = p_\sigma h p_\sigma^{-1}$. It follows that

$$N_{U(n)}(\bar{T}) = P_n \bar{T}, \text{ where } P_n = \{p_\sigma : \sigma \in \text{Sym}(n)\}.$$

$$\text{Thus, } W(U(n)) = P_n \bar{T} / \bar{T} \cong P_n \cong \text{Sym}(n).$$

$$\text{if } \sigma \in W(U(n)), \sigma \cdot \begin{bmatrix} z_1 & & 0 \\ & \ddots & \\ 0 & & z_n \end{bmatrix} = \begin{bmatrix} z_{\sigma(1)} & & 0 \\ & \ddots & \\ 0 & & z_{\sigma(n)} \end{bmatrix}.$$

$$(ii) \text{ } SU(n). \text{ Max's } T' = \left\{ \begin{bmatrix} z_1 & & 0 \\ & \ddots & \\ 0 & & z_n \end{bmatrix} : z_k \in \bar{\mathbb{C}}, \prod_{k=1}^n z_k = 1 \right\}$$

$$\text{Then } N_{SU(n)}(T') = N_{U(n)}(T) \cap SU(n)$$

$$= \left\{ p_\sigma h : \sigma \in \text{Sym}(n), h \in T, \underbrace{(\det p_\sigma)}_{\text{sgn } \sigma} (\det h) = 1 \right\}$$

if $\text{sgn } \sigma = -1$, and $h, h' \in T$, $\det h = -1 = \det h'$.

So if $p_\sigma h T' = p_\sigma h' T'$, then $p_\sigma T' = p_\sigma h' h^{-1} T'$.

and it follows $N_{SU(n)}(T') / T' \cong P_n \cong \text{Sym}(n)$.

Recall: $\Delta(\mathfrak{u}(2m)_{\mathbb{C}}) = \{ \pm(e_i - e_j), \pm(e_i + e_j) : 1 \leq i, j \leq m \}$

$$e_i(H_k) = \delta_{ik}, \quad H_k = E_{kk} - E_{k+m, k+m}$$

$$X_{ij} = E_{ij} - E_{i+m, j+m}, \quad \mathbb{C}X_{ij} = \mathfrak{h}(E_{2m})e_i - e_j$$

$i \neq j$

$$Y_{ij} = E_{ij+m} - E_{j, i+m}, \quad \mathbb{C}Y_{ij} = \begin{cases} \mathfrak{h}(E_{2m})e_i + e_j, & i < j \\ \mathfrak{h}(E_{2m})-e_i - e_j, & j < i \end{cases}$$

let us consider $N := N_{SO(E_{2m})}(T)$ and compute $W(SO(E_{2m}))$.

The conjugation action of N on T must preserve eigenvalues,

subject to restriction

- $\overline{z_k} = z_{k+m}, \quad k: 1, \dots, m$

- Otherwise only permutations are allowed

- have determinant 1.

$$\text{let } \tilde{S}_k = \sum_{\substack{j=1 \\ j \neq k}}^m (\bar{L}_{jj} - \bar{L}_{j+m, j+m}) + \bar{L}_{k, k+m} - \bar{L}_{k+m, k}, \text{ so}$$

$$\tilde{S}_k h_z \tilde{S}_k^{-1} = \begin{bmatrix} z_1 & & & & \\ & \ddots & & & \\ & & \bar{z}_k & & \\ & & & \ddots & \\ & & & & z_m \\ & & & & & \ddots \\ & & & & & & \bar{z}_1 \\ & & & & & & & \ddots \\ & & & & & & & & z_k \\ & & & & & & & & & \ddots \\ & & & & & & & & & & \bar{z}_m \end{bmatrix} \quad (\text{change } z_k \rightarrow \bar{z}_k, \bar{z}_k \rightarrow z_k)$$

$$\text{and } \det \tilde{S}_k = -1, \text{ so } \tilde{S}_k \in O(\mathbb{C}_{2m})$$

$$\text{let for } \sigma \in \text{Sym}(m), \text{ let } \tilde{P}_\sigma = \begin{pmatrix} P_\sigma & 0 \\ 0 & P_\sigma^T \end{pmatrix} - P_\sigma \text{ rotation matrix} \\ \in SO(2m)$$

$$\text{let } \tilde{P}_m = \{ \tilde{P}_\sigma : \sigma \in \text{Sym}(m) \} \in SO(\mathbb{C}_{2m})$$

$$\tilde{S} = \langle \tilde{S}_1, \dots, \tilde{S}_m \rangle \cap SO(\mathbb{C}_{2m})$$

$$= \{ \tilde{S}_1^{a_1} \dots \tilde{S}_m^{a_m} : a_m \in \{0, 1\}, 2 \mid a_1 + \dots + a_m \}$$

$$\cong \{ a \in \{-1, 1\}^m : a_1 \dots a_m = 1 \} := S$$

$$\cong \{-1, 1\}^{m-1}$$

Then (it is "easy" to check that)

$$N = \tilde{P}_m \tilde{S}^T$$

$$\Rightarrow W(SO(2m)) = W(SO(E_{2m})) \cong \tilde{P}_m \tilde{S}$$

$$\cong P_m S \quad , \quad P_m \text{ } m \times m \text{ permutation matrix}$$

$$\cong S \rtimes \text{Sym}(m)$$

↑ semidirect product

$$(a, \sigma)(a', \sigma') = (a(\sigma \cdot a'), \sigma\sigma')$$

Thus, $|W(SO(2m))| = 2^{m-1} m!$

Exercise: $W = (a, \sigma) \in S \rtimes \text{Sym}(m)$. then

$$W(e_i \pm e_j) = a_i e_{\sigma(i)} \pm a_j e_{\sigma(j)}$$

(iii). $SO(2m+1) \cong SO(E_{2m+1})$, we let

$$\tilde{S}_k = \sum_{\substack{j=1 \\ j \neq k}}^m (E_{jj} - E_{1+m, j+m}) + E_{k, k+m} - E_{k+m, k} - E_{2m+1, 2m+1}, \det \tilde{S}_k = 1.$$

it has extra roots $\pm e_k, k=1, \dots, m,$

$$\text{ad}(E_{\pm m+1})e_k = \mathbb{Q} E_k, \pm m+1$$

Here $N_{\text{SO}(2m+1)}(T) = \tilde{P}_m < \tilde{S}_1, \dots, \tilde{S}_k > T$

$$W(\text{SO}(2m+1)) \cong \{-1, 1\}^m \rtimes \text{Sym}(m), \text{ cardinality } 2^m m!$$

G -compact, connected matrix group, $\mathfrak{g} = \text{Lie}(G)$

T max'l torus, $\mathfrak{t} = \text{Lie}(T)$ Cartan subalgebra.

Recall: $\mathcal{U}(T) = \text{span}_{\mathbb{C}} \overline{T}, T \cong T^n, \overline{T} \cong \mathbb{Z}^n = \{\chi_{\mu} : \mu \in \mathbb{Z}^n\}$

$$\chi_{\mu}(z) = z^{\mu} = z_1^{\mu_1} \dots z_n^{\mu_n}$$

Define: $M_G(G) = \{f \in \mathcal{U}(G) : f(ghg^{-1}) = f(h), g, h \in G\}.$

$$= \text{span}_{\mathbb{C}} \{ \chi_{\pi} : \pi \in \hat{G} \}, \quad \chi_{\pi}(g) = \text{Tr}(\pi(g)), \quad g \in G.$$

$$\mathcal{M}_W(T) = \{ f \in \mathcal{M}(T) : f(wh) = f(h), \quad w \in W, \quad h \in T \}.$$

Corollary (to last Thm.)

The restriction map

$$f \mapsto f|_T : \mathcal{M}_G(G) \rightarrow \mathcal{M}_W(T)$$

is an injection.

Goal: identify $\{ \chi_{\pi} : \pi \in \hat{G} \}$ by identifying

$$\{ \chi_{\pi}|_T : \pi \in \hat{G} \} \subseteq \mathcal{M}_W(T)$$

Def'n: $A(t) = \text{span}_{\mathbb{C}} \{ e^{\lambda} : \lambda \in i\mathfrak{t}^* \}.$

$$(\lambda : \mathfrak{t}^* \rightarrow i\mathbb{R} = \mathfrak{u}(1) \text{ linear})$$

if $w \in W$, $f \in A(t)$, let $w \cdot f(H) = f(w^{-1} \cdot H)$

$$\text{So } w \cdot e^\lambda = e^{w \cdot \lambda}$$

let $A_W(t) = \{f \in A(t) : w \cdot f = f \text{ for } w \in W\}$ - W -symmetric

$A_W^-(t) = \{f \in A(t) : w \cdot f = (\det w) f \text{ for } w \in W\}$ - W -alternating.

"integral form"

$$\Gamma(t) = \{\lambda \in t^* : \lambda(k) \in 2\pi i \mathbb{Z}\}, \quad k = \ker \exp,$$

$\exp: t \rightarrow T$ is a covering map $k \cong \mathbb{Z}^n$.

Notes: (i) $A(t)$ is an algebra of functions ($e^\lambda e^\mu = e^{\lambda+\mu}$)

(ii) $A_W(t)$ is a subalgebra of $A(t)$

$A_W^-(t) A_W^-(t) \subseteq A_W(t)$. ($\det w = \pm 1$, $H \mapsto w \cdot H$ is linear.

$\det w$ is the determinant of the operator

$$\text{iii) If } \lambda \in I(\mathfrak{t}), e^\lambda \in \tilde{M}(T) \\ \uparrow \\ T \cong \mathfrak{t}/k.$$

On inner product: Recall that for compact matrix G .

$G \in U(n)$, $\text{Lie}(G) = \mathfrak{g} \in u(n)$, and $u(n)$ admits a real inner product, so $(\text{Ad}_G X, \text{Ad}_G Y) = (X, Y)$, $X, Y \in \mathfrak{g}$.

i.e., $\text{Ad}(G) \in SO(\mathfrak{g})$. $\text{ad}(\mathfrak{g}) \in \mathfrak{so}(\mathfrak{g})$. Hence:

$$\text{(i) if } \alpha \in \Delta(\mathfrak{g}_{\mathbb{C}}) \Rightarrow -\alpha \in \Delta(\mathfrak{g}_{\mathbb{C}})$$

$$(\alpha: \mathfrak{t} \rightarrow u(1) = i\mathbb{R}, \text{ i.e., } \alpha \in i\mathfrak{t}^*)$$

(ii) given $\lambda \in \mathfrak{t}^*$, there is H_λ in $\mathfrak{t}$ so

$$\lambda(H) = (H, H_\lambda)$$

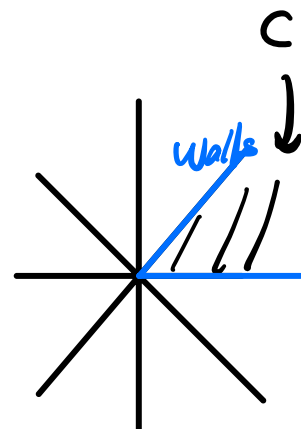
Consequence, $\alpha \in i\mathfrak{t}^*$, $\alpha(H) = i(H, H\alpha)$

($H\alpha$ suitably chosen for $-i\alpha \in \mathfrak{t}^*$)

Def'n: A Weyl chamber C of $\mathfrak{t}$ is a connected component of

$$\mathfrak{t} \setminus \bigcup_{\alpha \in \Delta(\mathfrak{g}_{\mathbb{C}})} H_{\alpha}^{\perp} = \mathfrak{t} \setminus \bigcup_{\alpha \in \Delta(\mathfrak{g}_{\mathbb{C}})} \ker \alpha$$

orthogonal complement
i.e., hyperplane



The walls of C are $H_{\alpha}^{\perp} \cap \overline{C}$

For $\alpha \in \Delta(\mathfrak{g}_{\mathbb{C}})$, define $r_{\alpha} \in O(\mathfrak{t})$ (orthogonal op. on $\mathfrak{t}$)

$$r_{\alpha}(H) = H - 2 \underbrace{\frac{(H, H\alpha)}{(H\alpha, H\alpha)}}_{\text{projection onto } \mathbb{R}H\alpha} H\alpha, \quad r_{\alpha} = r_{\alpha}^T = r_{\alpha}^{-1}.$$

Hence r_α fixes $H_\alpha^\perp = \ker \alpha$, and is reflection
through this hyperplane.

Black Box: For each α , there is w_α in W (Weyl group)

so $r_\alpha(H) = w_\alpha \cdot H$, and $w_\alpha^{-1} = w_\alpha$. Furthermore,

$\langle w_\alpha : \alpha \in \Delta(\mathfrak{g}_{\mathbb{C}}) \rangle = W$. Finally, W acts faithfully
and transitively on the set of Weyl chambers.

(Hence, $\det w_\alpha = \pm 1$).

Note: There is an abstract notion of reflection groups.

"Weyl group", which arises in classification scheme

of simple compact Lie groups.
 $\mathfrak{g}$ simple

Def'n: A subset $\Delta^+ \subseteq \Delta(\mathfrak{g}_{\mathbb{C}})$ is a set of simple roots

if $\Delta(\mathfrak{g}_{\mathbb{C}}) \cong \pm \text{span}_{\mathbb{N}_0} \Delta^+$, $\mathbb{N}_0 = \{0\} \cup \mathbb{N}$

(each root is an integer comb. of Δ^+),

If C is a Weyl chamber, then

$$\Delta^+ = \Delta^+_C = \{ \alpha \in \Delta(\mathfrak{g}_{\mathbb{C}}) : (H, H_{\alpha}) > 0 \text{ for } H \in C \}$$

Ex: $U(n)$, $SU(n)$

$$\Delta^+ = \{ e_i - e_j : 1 \leq i < j \leq n \} \subseteq \{ \pm(e_i - e_j) : 1 \leq i < j \leq n \} = \Delta(\mathfrak{u}(n)_{\mathbb{C}}).$$

corresponds to $C = \left\{ \begin{bmatrix} i\theta_1 & & 0 \\ & \ddots & \\ 0 & & -i\theta_n \end{bmatrix} : \theta_1 > \dots > \theta_n \in \mathbb{R} \right\}$

$$\begin{aligned} SO(2m) : \Delta^+ &= \{ e_i - e_j, e_i + e_j : 1 \leq i < j \leq m \} \subseteq \{ \pm(e_i - e_j), \pm(e_i + e_j) : 1 \leq i < j \leq m \} \\ &= \Delta(\mathfrak{so}(2m)_{\mathbb{C}}) \end{aligned}$$

$$C = \left\{ \begin{bmatrix} -i\theta_1 & & & \\ & \ddots & & \\ & & -i\theta_m & \\ 0 & & & 0 \\ & & -i\theta_1 & \\ & & & \ddots & \\ & & & & -i\theta_m \end{bmatrix} : \theta_1 > \dots > \theta_m > 0, m \in \mathbb{R} \right\}$$

$$\mathcal{I}O(E_m)$$

$$SO(2m+1) : \Delta^+ = \{e_i - e_j, e_i + e_j, e_k : 1 \leq i < j \leq m, k=1, \dots, m\}$$

Notation: let $\rho = \frac{1}{2} \sum_{\alpha \in \Delta^+} \alpha \in i\mathfrak{t}^*$, so $2\rho \in \mathcal{I}(\mathfrak{t})$.

and $H_\rho = \frac{1}{2} \sum_{\alpha \in \Delta^+} H_\alpha \in C$.

Exercise: $U(n), SU(n) : \rho = \frac{1}{2} \sum_{k=1}^n (n-k+1) e_k$

$$SO(2n) : \rho = \sum_{k=1}^n (n-k) e_k \in \mathcal{I}(\mathfrak{t})$$

$$SO(2m+1) : \rho = \sum_{k=1}^m (n-k+\frac{1}{2}) e_k.$$

(Recall that $\mathcal{I}(\mathfrak{t}) = \{\lambda \in i\mathfrak{t}^* : \lambda(k) \in 2\pi i \mathbb{Z}\}, k = \ker(\exp),$

$\exp: \mathfrak{t} \rightarrow T$ is a covering map; $\alpha = d\chi \in \mathcal{I}(\mathfrak{t}), \chi \in \widehat{T}$)

Basic Alternating functions:

$$\text{let } A: \mathfrak{t}^* \rightarrow A_{\mathfrak{t}}(\mathfrak{t})$$

$$A(\lambda) = \sum_{w \in W} (\text{det } w) e^{w \cdot \lambda}$$

Notes: For $\lambda \in \mathfrak{t}^*$, $A(\lambda) = 0$

$$\Leftrightarrow w \cdot \lambda = \lambda \text{ for some } w \in W \setminus \{1\}$$

$$\Leftrightarrow H_\lambda \in \bigcup_{\alpha \in \Delta(g_c)} H_\alpha^\perp \quad (\text{walls})$$

Notation: Define $f \in A(\mathfrak{t})$, $f(H) = \prod_{\alpha \in \Delta^+} (e^{\frac{1}{2}\alpha(H)} - e^{-\frac{1}{2}\alpha(H)})$, $H \in \mathfrak{t}$

Notes: i) $f(H) = 0 \Leftrightarrow e^{\alpha(H)} = 1$ for all $\alpha \in \Delta^+$

$$\Leftrightarrow H \in \bigcup_{\alpha \in \Delta^+} \alpha^{-1}(2\pi i \mathbb{Z})$$

$$\text{we denote } \mathfrak{t}_{\text{reg}} = \mathfrak{t} \setminus \bigcup_{\alpha \in \Delta^+} \alpha^{-1}(2\pi i \mathbb{Z})$$

"regular"

$\uparrow$ could take $\alpha \in \Delta(g_c)$

$$\text{ii)} \quad \delta = e^{\delta} \underbrace{\prod_{\alpha \in \delta^+} (1 - e^{-\alpha})}_{\substack{\text{modulus} \\ 1}} \in \mathcal{I}(\mathfrak{t})$$

$$\text{(iii)} \quad (\text{Harder}) \quad A(\rho) = \delta.$$

Prop 4: (Schur functions)

If $\lambda \in \mathcal{I}(\mathfrak{t})$, then

$$S_\lambda = \frac{A(\rho + \lambda)}{\delta} \quad (\text{makes sense on } \mathfrak{t}_{\text{reg}})$$

extends to a cts fun on $\mathfrak{t}$, and in fact,

S_λ is constant on additive cosets of

$$(\mathfrak{k} = \ker(\exp) : \exp : \mathfrak{t} \rightarrow T)$$

Hence $S_\lambda = \tilde{S}_\lambda \circ \exp$, $\tilde{S}_\lambda \in \mathcal{M}_w(T)$.

Thm: let $I^+(t) = \{\lambda \in I(t) : (H, H_\alpha) \geq 0 \text{ for } H \in C\}$
↑
wely chamber

Then: • each $S_\lambda, \lambda \in I^+(\lambda)$ is $\chi_\pi|_T$

for some $\pi \in \hat{G}$, and all such $\chi_\pi|_T$

are realized this way:

$$\lambda \mapsto \pi_\lambda : I^+(t) \rightarrow \hat{G}$$

• (dimension formula)

$$\dim \underbrace{V_{\pi_\lambda}}_{\substack{\text{space on which} \\ \pi_\lambda \text{ acts}}} = \text{Tr}_\lambda(I_{\pi_\lambda}) = \chi_{\pi_\lambda}(I) = S_\lambda(I)$$

π_λ acts

$$= \prod_{\alpha \in \Delta^+} \frac{(H_\alpha, H_{\lambda+\rho})}{(H_\alpha, H_\rho)},$$

Recall: $\rho = \frac{1}{2} \sum_{\alpha \in \Delta^+} \alpha$, $\delta = \prod_{\alpha \in \Delta^+} (e^{\frac{\alpha}{2}} - e^{-\frac{\alpha}{2}})$. $A(\rho) = \delta$

Idea: $\underbrace{\{A(\lambda) : \lambda \in I^+(t)\}}_{\sum_{w \in W} (\det w) e^{w \cdot \lambda}}$ descends to an orthonormal set
 constant on cosets of k in $M_W(T)$
 So by 1st iso-thm, define fns on T .

$A_w(t)$ is freely gen. as a free abelian gp ($\mathbb{Z}$ -module)
 over $\{A(\lambda) : H_\lambda \in \mathbb{C}\}$.

Weyl Integral Formula: let $f \in \mathcal{C}(G)$, then

$$\underbrace{\int_G f}_{\text{left/right-inv int.}} = \frac{1}{|W|} \int_T \det \left(\text{Ad}(h^{-1})|_{\mathfrak{g}^\perp} - I_{\mathfrak{g}^\perp} \right) \underbrace{\int_G f(ghg^{-1}) dg}_{\text{orth. Compl.}} d'$$

$\int_G 1 = 1$ $\int_{G/T} f(ghg^{-1}) dg$

$$(f \text{ is a class function, } |W| \int_G f = \int_G \det(\text{Ad}(h^{-1}) - I)|_{\mathfrak{t}^\perp})$$

What's going on? $\mathcal{P}: (G/T, T) \rightarrow G$

$$\mathcal{P}(gT, h) = ghg^{-1} \quad - \text{ parametrizes } G$$

covering map with degree $|W|$.

$$\det(\text{Ad}(h^{-1})|_{\mathfrak{t}^\perp} - I_{\mathfrak{t}^\perp}) = \det(J_{\mathcal{P}}(gT, h))$$

$$\mathfrak{g}_\mathbb{C} = \mathfrak{k}_\mathbb{C} \oplus \bigoplus_{\alpha \in \Delta(\mathfrak{g}_\mathbb{C})} \mathfrak{g}_\alpha, \quad \begin{array}{l} \text{Ad}(T) \in \text{SO}(\mathfrak{g}) \\ \quad \quad \quad \subseteq \text{SO}(\mathfrak{k}^\perp) \end{array}$$

each $\alpha \in \Delta(\mathfrak{g}_\mathbb{C})$ admits $-\alpha$ in $\Delta(\mathfrak{g}_\mathbb{C})$

$\Rightarrow$ eigenvalues of $\text{Ad}(\exp(-H))|_{\mathfrak{t}^\perp} - I_{\mathfrak{t}^\perp}$ are

$$e^{\pm \alpha(H)} - 1, \quad \alpha \in \Delta^+ \text{ (realizes half of roots)}$$

$\exp(-H)$ eigenvalues are $\pm \alpha(H)$.

$$\begin{aligned} \text{So } \det(\text{Ad}(\exp(-H))|_{\mathfrak{g}_i} - I_{\mathfrak{g}_i}) &= \prod_{\alpha \in \mathfrak{g}_i} (e^{\alpha(H)} - 1)(e^{-\alpha(H)} - 1) \\ &= f \cdot \bar{f} \end{aligned}$$

Now let $\pi \in \hat{G}$. Then $\chi_\pi = \text{Tr} \circ \pi \in M_w(T)$,

so $\chi_\pi \circ \exp \in A_w(t)$, and hence

$$\begin{array}{c} (\chi_\pi \circ \exp) f \in A_w(t) \\ \uparrow \\ A_w(t) \end{array}$$

By Schur orthogonality,

$$1 = \int_G \chi_\pi \overline{\chi_\pi} = \frac{1}{|w|} \int_I (\chi_\pi f) \overline{(\chi_\pi f)}$$

if $\pi' \neq \pi$ in $\hat{G}$,

$$0 = \int_G \chi_\pi \chi_{\pi'} = \frac{1}{|w|} \int_I (\chi_\pi f) \overline{(\chi_{\pi'} f)}$$

Reduction steps: one shows that

$$(X_{\tau_1} \circ \exp) f = A(\gamma_{\tau_1}), \gamma_{\tau_1} \in \overline{C},$$

And then, one finds $\gamma_{\tau_1} = \rho + \lambda$, $\lambda \in \mathcal{I}^+(\mathcal{L})$.